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Applied Analytics Project - Week 9 Assignment

Validation Error for All Models

Multiple models were evaluated to compare performance. The most basic and worst performing model was the initial logistic regression. The more complex models are listed below. The validation error (1-accuracy) for each model is as follows:

Model	Validation Accuracy	Validation Error
adaBoost	0.809524	0.190476
randomForest	0.793651	0.206349
extraTrees	0.777778	0.222222
gradientBoosting	0.809524	0.190476
histGradientBoosting	0.809524	0.190476
svc	0.730159	0.269841
mlp	0.793651	0.206349
xgboost	0.809524	0.190476

The best validation error was 0.190476. This result came from multiple models, including the adaBoost, gradientBoosting, histGradientBoosting, and xgboost models.

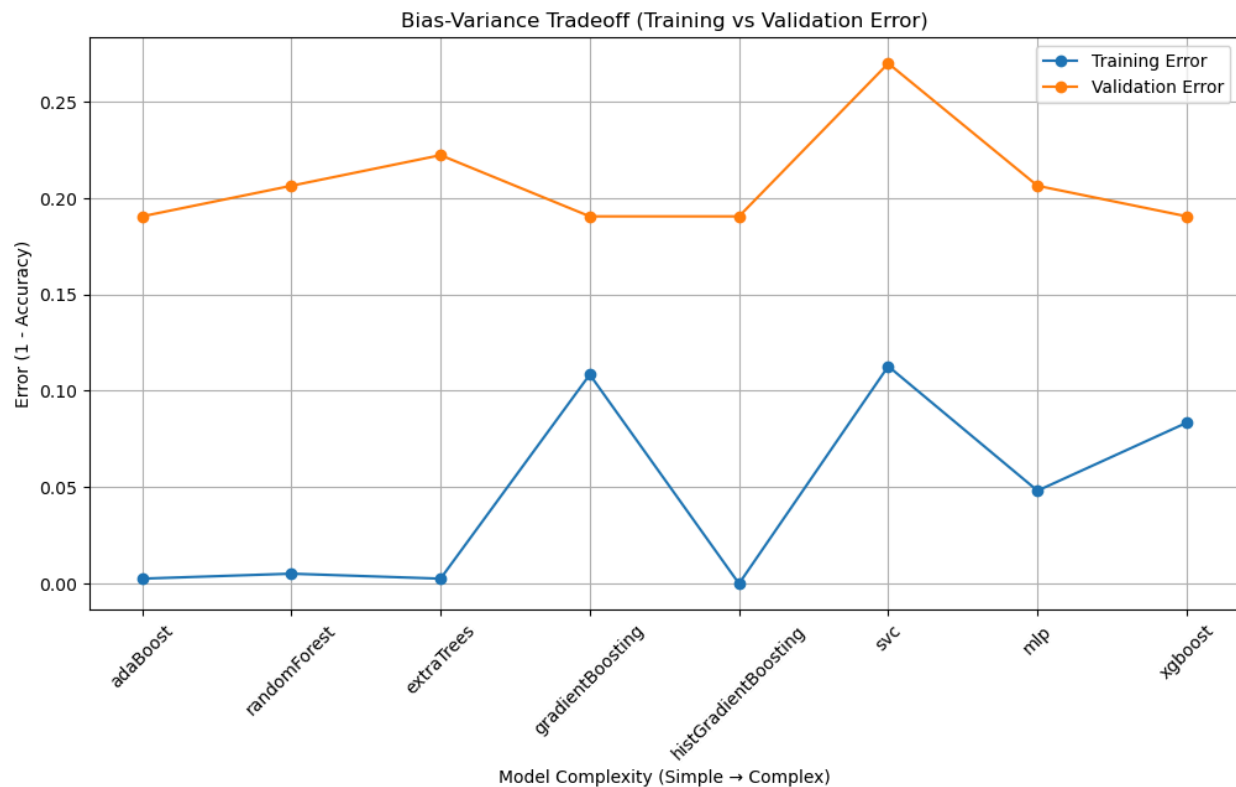
Selection of the Final Winning Model

The final winning model based on the best validation performance is the xgboost model configuration 18. This model was selected as the winning model because it had the highest validation accuracy (0.809524) and therefore the highest validation error (0.190476). It also has a high recall rate of 0.975. Although a few models had these same validation metrics, xgboost models reduce overfitting risk by using a regularized boosting algorithm. They also

capture nonlinear relationships well and use low learning rate and subsampling to improve generalization.

Bias-Variance Tradeoff

The below chart shows the bias-variance tradeoff between the different models.



The bias-variance tradeoff chart shows how the simpler models on the left side (adaBoost, randomForest, extraTrees) have higher bias. This is evident from the very low training errors, which are near 0, and the higher validation error, which hovers around 0.20. This tells us that these models are too simple and do not fully capture the patterns of the dataset. The high bias indicates underfitting. The middle region (gradientBoosting, histGradientBoosting) has the lowest validation error, which is optimal. The training error is still low, but not low enough to indicate underfitting. These models capture data patterns well and show a good balance between bias and variance. The right side of the chart (svc, mlp, xgboost) shows higher training and validation errors. This shows that these models do not generalize well and indicate overfitting.

Final Model Test Dataset Performance

The test dataset uses data from 2023-2025 seasons. The evaluation metrics from the test data are as follows:

Accuracy	0.6953
Precision	0.7091
Recall	0.9176

In addition, the below table shows the actual versus predicted results for the test dataset.

	Predicted 0	Predicted 1
Actual 0	11	32
Actual 1	7	78

The test accuracy of 0.6953 is lower than the validation accuracy of 0.8095. This indicates that some overfitting occurred and the model did not fully generalize. Recall is still high at 0.9176, indicating that the model is good at identifying wins. Precision is moderate, so there might be some false positives. Given that recall is good, the model is acceptable. However, since accuracy is a bit low, the model is not ideal for balanced prediction and there is room for improvement.

Training vs. Validation vs. Test Performance

	Accuracy	Precision	Recall
Training	0.9165	0.8944	0.9936
Validation	0.8095	0.78	0.975
Test	0.6953	0.7091	0.9176

These results show that the training set performs the best, the validation set performs moderately, and the test set performs the worst. This is expected and indicates some

overfitting between the training and validation sets. The drop in test performance indicates some generalization issues.